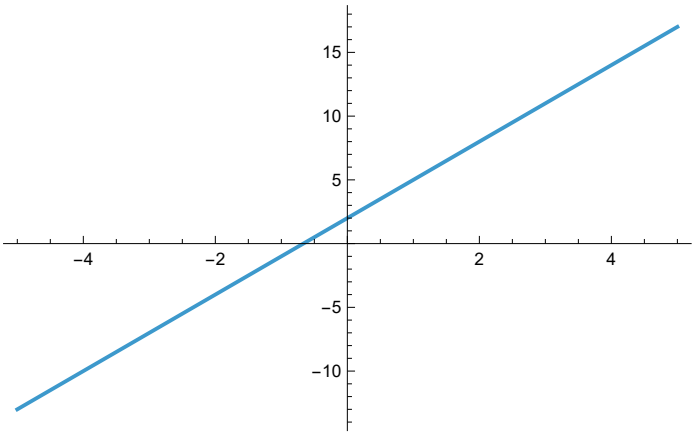
```
(* MAT120 Lecture 1 Lab Notes *)

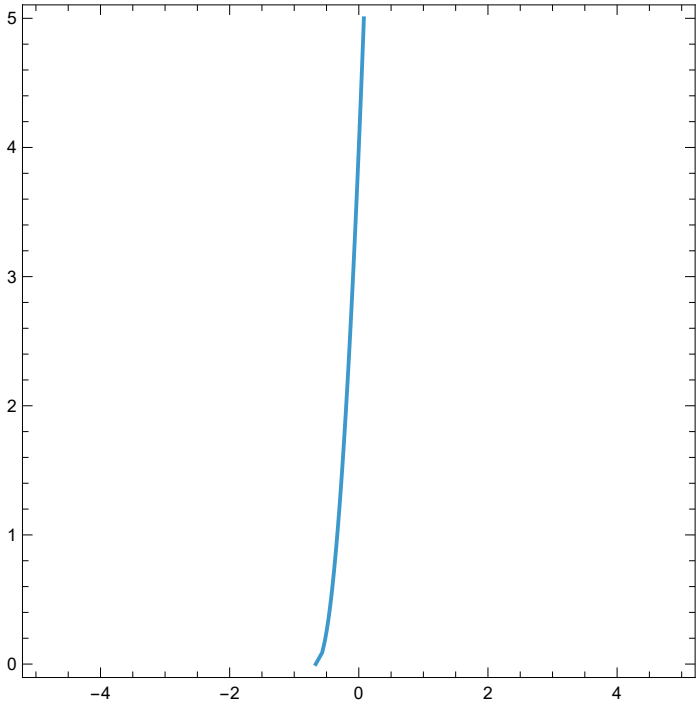
Plot[3 x + 2, {x, -5, 5}]
```

Out[•]=



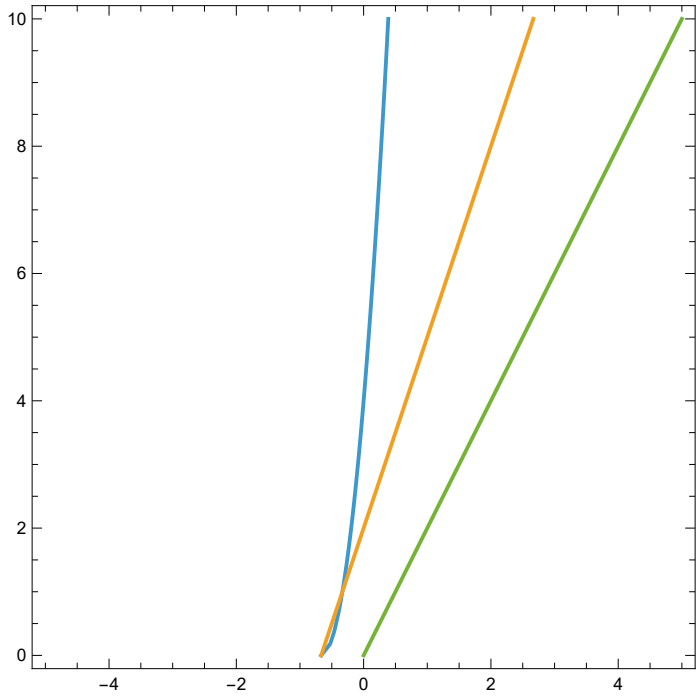

```
ContourPlot[ √y == 3 x + 2, {x, -5, 5}, {y, 0, 5}]
```

Out[•]=



```
(* y = 3x + 2, y = 3x and √y =3x+2 - show these three functions using Mathematica. *)
ContourPlot[{√y == 3 x + 2, y == 3 x + 2, y == 2 x}, {x, -5, 5}, {y, 0, 10}]
```

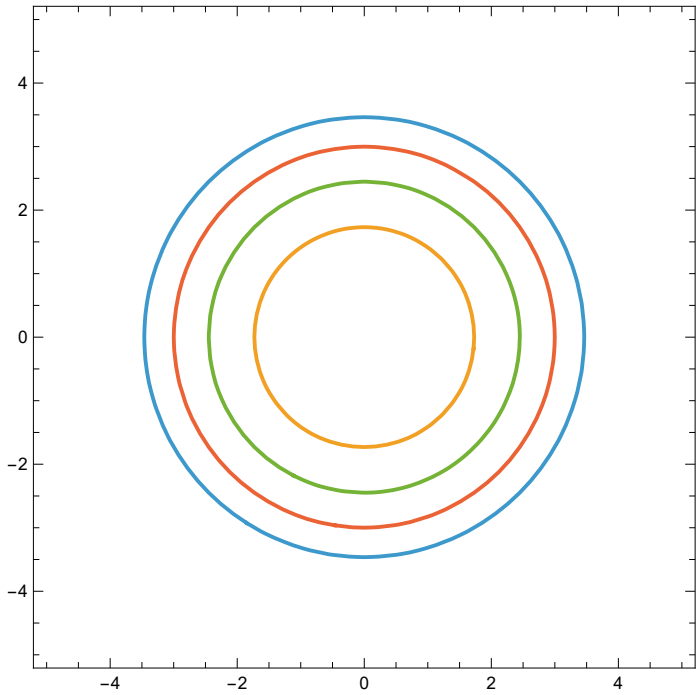
Out[•]=



```
(* x^2 + y^2 = 16, graph this function using Mathematica.

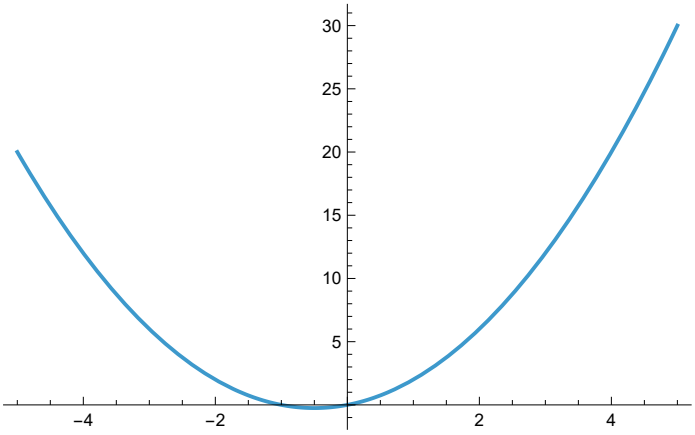
  NOTE: for circle, parabola, hyperbola, ellipse, the range must be same *)
ContourPlot[{x^2 + y^2 == 12, x^2 + y^2 == 3, x^2 + y^2 == 6, x^2 + y^2 == 9},
 {x, -5, 5}, {y, -5, 5}]
```

Out[•]=




```
(* Plot x^n + x using Mathematica command. n = 2 because yes. *)
Plot[x^2 + x, {x, -5, 5}]
```

Out[•]=



In[•]:=
```
ContourPlot[ √y == x^4 + x, {x, -2, 2}, {y, -2 , 2}]
```

Out[•]=

